

Shubham Shah

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Summary

Software Engineer with **5+ years of experience** developing enterprise applications using **Java, Spring Boot, and Angular** in microservices and modular front-end architectures. Skilled in building **secure APIs** with OAuth2, JWT, and Spring Security, and implementing **event-driven systems** with Kafka, schema validation, and retry handling. Proficient in both **SQL-PostgreSQL, MySQL, Oracle** and **NoSQL-MongoDB, DynamoDB** databases with a focus on performance optimization and compliance logging. Experienced in **AWS cloud deployments** (ECS, Lambda, S3, EC2) with Docker and CI/CD pipelines using Jenkins and GitHub Actions. Adept at **monitoring and troubleshooting** production systems with Splunk, OpenTelemetry, and CloudWatch to ensure reliability and resilience. Strong advocate of **Agile practices**, recognized for clear communication, cross-team collaboration, and taking ownership of deployments, testing, and continuous improvements.

Technical Tools

Backend Development: Java 8, Java 11, Java 17, Spring Boot, Spring MVC, Spring Security, Spring Profiles, RESTful APIs, Servlets, DTO Mapping

Frontend Development: Angular 8/10/14/18, TypeScript, JavaScript, RxJS, Reactive Forms, Angular Interceptors, Material Design

Databases: PostgreSQL, MySQL, Oracle, MongoDB, Amazon RDS, NoSQL, SQL, TTL Indexing

Messaging & Event Streaming: Kafka, Schema Validation, Topic Monitoring, Offset Handling, Retry Topics, Embedded Kafka

DevOps & Deployment: Docker, Kubernetes, Jenkins, CI/CD Pipelines, WAR Deployment, Bash Scripts, GitHub Actions, Manual Config

Cloud & Infrastructure: AWS ECS, Lambda, S3, EC2, Parameter Store, Secrets Manager, CloudWatch

Security & Auth: OAuth2, JWT, RBAC, Spring Security Filters, Role-Based Access Control, Session-Based Authentication

Testing: JUnit, Mockito, Embedded Kafka, Integration Testing, Load Testing, K6

Monitoring & Logging: Splunk, OpenTelemetry, Log4J, CloudWatch Dashboards, MDC Injection

Build & Configuration: Maven, YAML, Jenkins Jobs, Environment Variables, Property Files

Legacy & Integration: JSP, JSTL, Struts, SOAP, WSDL, JAXB, XML Parsing, DOM Parser, jQuery, AJAX

Process & Collaboration: Agile, Sprint Planning, Retrospectives, Jira, QA Handoffs, Technical Communication

Experiences

Church Mutual Insurance | Software Engineer

Nov 2024 – Current | Merrill, WI, USA

Project: Policy Document Tracker

Built and supported a **backend service and Angular-based interface** for managing policy document lifecycle, including version history, user acknowledgments, and expiration notices. The project addressed the need for internal teams to track compliance documents tied to specific coverage types and policyholders. Used **event-driven triggers** to update document status and display real-time badges on the dashboard.

Responsibilities:

- Developed secure **Java 17** microservices using **Spring Boot** with event-driven patterns and fallback logic through circuit breaker in a real-time notification engine.
- Designed **Angular 18** front-end modules with lazy loading, route guards, and **Angular Interceptors** to protect routes and manage token expiration gracefully.
- Built **RESTful APIs** in Spring Boot for user registration and profile access, enabled actuator endpoints to monitor uptime and expose application health metrics.
- Configured **Spring Profiles** in YAML for environment isolation and externalized values using **AWS Parameter Store** to support ECS and EC2 runtime environments.
- Created **Kafka** producers with schema validation using Avro and enabled topic monitoring for failed payloads and consumer lag analytics.
- Integrated authentication and authorization using **OAuth2**, JWT, and RBAC with custom **Spring Security filters** for scoped endpoint access.
- Used **TypeScript** to develop shared services and reactive forms for onboarding components inside Angular, supporting field-level validation and role-based toggles.
- Designed **PostgreSQL** schema for ticket tracking features and optimized SQL joins and indexes for dashboard and SLA monitoring modules.
- Deployed microservices on **AWS ECS** and created **Docker** images with Spring Boot configurations using **Secrets Manager** for encrypted credential injection.
- Created serverless **AWS Lambdas** triggered by **S3** file uploads to extract metadata and initiate document classification workflows.
- Integrated **JUnit** and **Mockito** tests covering controller responses, service logic, and Kafka message retries, using embedded Kafka for local validations.
- Managed production **deployment** using rolling strategies in Jenkins **CI/CD**, coordinated rollback plans, and tracked code smells using **SonarQube** reports.
- Configured **OpenTelemetry** to trace Kafka flows and linked them to **Splunk** dashboards, improving traceability in distributed microservices.
- Used **JavaScript** inside Angular widgets to render live charts and bind keyboard shortcuts for frequent actions in tabular views.
- Contributed to internal GenAI PoC for backend NLP classification using **AWS SageMaker APIs** and integrated tokenized input workflows into event stream processing.
- Used **MongoDB** to persist audit events for compliance logging and configured capped collections for periodic expiration.
- Participated in daily **Agile** scrum meetings and sprint planning sessions with product, QA, and dev teams for technical clarity and **communication**.
- Resolved production issues using **Linux** tools like grep, journalctl, and tail for log analysis and restarted failing containers in ECS via CLI.
- Built dynamic forms using Angular with **Material Design** libraries and performed accessibility testing across multiple device resolutions and breakpoints.

Technological Environment::

Java 17, Spring Boot, Spring Profiles, Spring Actuator, Spring Security, OAuth2, JWT, RBAC, RESTful APIs, Angular 18, TypeScript, JavaScript, Angular Interceptors, Lazy Loading, Material Design, PostgreSQL, MongoDB, Kafka, Schema Validation, Topic Monitoring, AWS ECS, AWS Lambda, AWS EC2, AWS S3, Parameter Store, Secrets Manager, Docker, CI/CD, JUnit, Mockito, SonarQube, OpenTelemetry, Splunk, Agile, Linux, YAML, GenAI, SageMaker, Reactive Forms, Avro, Circuit Breaker, Communication

Project: Investor Insights Dashboard

Built a **client-facing analytics platform** to present bond performance metrics, issuer risk categories, and historical rating trends to institutional investors. Developed backend **Spring Boot services** to aggregate and serve rating data through **REST APIs**, incorporating pagination and filtering for efficient querying. Created **interactive Angular components** for visualizing data via charts and sortable grids, allowing users to compare issuers across regions and sectors.

Responsibilities:

- Refactored legacy services into modular **Java 11** components by applying interface segregation and DTO layering to support backward compatibility in Spring Boot applications.
- Created shared components in **Angular 14** using **RxJS** subjects to enable state communication and isolate component-level subscriptions for better reusability.
- Applied **Spring Boot** validation using DTOs and constraint annotations to enforce structured payload handling and reduce boilerplate across controller layers.
- Developed **RESTful APIs** with header-based routing logic and configured route-based version control in Spring Gateway for internal client requests.
- Created table structures in **PostgreSQL** to manage record logs and optimized filtering logic using partial indexes and time-bound range scans.
- Used **TypeScript** to build shared utility methods and implemented dynamic validators with rule-based form behavior mapped from backend configurations.
- Implemented **Kafka** retry strategies using custom backoff intervals and topic separation for error tracing and bad data quarantine.
- Designed service orchestration using standardized configs and API templates to handle **microservices** version updates without breaking downstream consumers.
- Developed reactive UI forms in **Angular** with lazy-loaded feature routes to reduce initial load size and improve performance in low-bandwidth scenarios.
- Modeled **NoSQL** data using MongoDB for distributed token state and improved query access patterns using compound indexes and TTL expiration policies.
- Processed incoming payloads with **Java 11** streams and parallel collectors, then dispatched cleaned records to Kafka producers for background consumption.
- Created Jenkins CI/CD pipelines with job-level quality gates, then deployed services using **Docker** with runtime ENV configuration and custom tagging logic.
- Deployed Spring Boot microservices on **Kubernetes** clusters by using **Helm charts** and ConfigMaps.
- Configured Secrets for environment variables in order to support secure and consistent application rollouts.
- Configured **Horizontal Pod Autoscaler (HPA)** with liveness probes and readiness probes to scale pods automatically.
- Deployed **AWS Lambda** functions to archive generated files into **S3**, connected output logs to CloudWatch, and sent completion events to downstream queues.
- Set up metric dashboards in **CloudWatch** and configured Splunk alerts using JSON log enrichment and OpenTelemetry trace correlation.
- Wrote functional tests using **JUnit** and **Mockito** to verify Kafka retry behavior, fallback flows, and error propagation through middleware filters.

- Collaborated in **Agile** grooming, conducted test case reviews with QA teams, and shared technical blockers during daily calls for effective **communication**.

Technological Environment::

Java 11, Spring Boot, DTO Mapping, Request Validation, RESTful APIs, Angular 14, RxJS, Lazy Loading, TypeScript, JavaScript, PostgreSQL, MongoDB, NoSQL, Microservices, Service Orchestration, Kafka, Retry Handling, Topic Setup, Docker, Jenkins, CI/CD, Deployment, Kubernetes, AWS Lambda, AWS S3, AWS EC2, CloudWatch, Splunk, JUnit, Mockito, Agile, Communication, OpenTelemetry, JSON Logs, Header-Based Routing, ENV Config

Leapfrog Technology | Java Developer

Dec 2019 – May 2023 | Seattle, Washington, USA

Project: Client Billing Portal

Developed a **web-based billing platform** for external clients to download invoices, track payment history, and raise disputes linked to service contracts. Built **Spring MVC controllers** with form-based login and integrated backend services to fetch invoice data from **Oracle procedures**.

Responsibilities:

- Developed **Java 8** backend services using **Spring MVC** and Servlet configuration to handle internal APIs, parameter bindings, and form-based login flows.
- Created dynamic web pages using JSP, JSTL, and EL expressions, backed by controller actions in a tightly coupled **MVC** architecture.
- Used **Angular 10** reactive forms and custom validators to implement conditional field logic and reused form structures across modules using shared services.
- Built internal APIs as **RESTful APIs** using **Spring Boot**, applied bean validation annotations, and connected with frontends through JSON-based data exchange.
- Created **Kafka** consumers with offset tracking and retries by storing delivery status in PostgreSQL, and implemented basic monitoring using partition lag metrics.
- Designed legacy data structures in **Oracle** for reference tables and transactional reports using stored procedures and foreign key-based access control.
- Modeled **MongoDB** collections for temporary storage of caching metadata and configured TTL policies for periodic cleanup of expired documents.
- Integrated third-party SOAP services using JAX-WS, consumed WSDL definitions, and performed schema validation using **XML** schemas before parsing payloads.
- Implemented UI logic using **JavaScript** and jQuery for modal interactions, dynamic select dropdowns, and content updates using AJAX for a more responsive experience.
- Handled manual **deployment** through WAR packaging into Apache Tomcat with Bash-based shell scripts for config switching and environment-specific setup.
- Authenticated users through legacy session-based management and encrypted token storage using database-backed security filters for restricted areas.
- Used **JUnit** and **Mockito** to test controller logic and helper utilities, wrote stubs for SOAP endpoints, and validated schema-mapped response parsing.
- Logged application behavior using **Log4J** with rolling file appenders and property-based log level control for tracking issues in legacy workflows.
- Participated in sprint ceremonies under the **Agile** model, collaborated with QA during regression windows, and maintained active **communication** during UAT reviews.

- Configured **Jenkins** jobs for CI pipelines to package builds and deploy to on-prem servers, added notification triggers for successful artifact publishing.
- Parsed nested **XML** payloads using DOM parser to extract keys from multi-level SOAP responses and log rejected entries for audit compliance.

Technological Environment::

Java 8, Spring, Spring Boot, Spring MVC, JSP, JSTL, Servlets, SOAP, RESTful APIs, XML, WSDL, JAXB, DOM Parser, Log4J, Oracle, PostgreSQL, MongoDB, MySQL, SQL, NoSQL, Kafka, Offset Tracking, JUnit, Mockito, Angular 10, Reactive Forms, RxJS, TypeScript, JavaScript, jQuery, AJAX, WAR Deployment, Apache Tomcat, Bash Scripts, Jenkins, CI/CD, Agile, Communication, Manual Configuration, Legacy Auth

Education

Clark University
Master Of Science In Computer Science

Vellore Institute of Technology
Bachelors in Computer Science and Engineering